



UNIVERSITY OF PERADENIYA
DEPARTMENT OF STATISTICS AND COMPUTER SCIENCE

STA1023/MAT1063- TUTORIAL 4

1. The MGF of Binomial random variable is $M_x(t) = [1 + \theta(e^t - 1)]^n$. Use it to derive the mean and the variance of $X \sim \text{Bin}(n, \theta)$.
2. If a fair coin is tossed at random five independent times, find the conditional probability of five heads given that there are at least 4 heads.
3. A fair coin is tossed 7 times. What is the probability that 6 will appear five or more times?
4. During a particular period, a university's information technology office received 20 service orders for problems with printers of which 8 were laser printers and 12 were inkjet models. A sample of 5 of these service orders is to be selected for inclusion in a customer satisfaction survey. If 5 service orders are selected randomly, what is the probability that exactly two of them were inject printers?
5. For a basketball player, the probability of making a free throw is 0.7 during a season.
 - a) What is the probability that he makes 3 free throws out of 8 shots?
 - b) What is the probability that he makes his 3rd free throw on his 5th shot?
 - c) What is the probability that he makes his 1st free throw on his 4th shot?
6. Let X be the number of typos on a printed page with a mean of 3 typos per page. Find
 - a) $P(X > 4)$
 - b) $P(X = 4)$
 - c) Probability that three randomly selected pages would have more than 8 typos.
7. An experiment of drawing a random card from an ordinary playing cards deck is done with replacing it back. This was done ten times. Find the probability of getting 2 spades, 3 diamond, 3 clubs and 2 hearts.
8. During a certain month, a company sold 10,000 new watches. Past experience indicates that the probability that a new watch will need repair during its warranty period is 0.002. Compute (Using Poisson Approximation), the probability that:
 - a) zero watches will need warranty work.
 - b) no more than 5 watches will need warranty work.
 - c) no more than 10 watches will need warranty work.
 - d) no more than 20 watches will need warranty work.

Submit On or Before September 14th, 2026 (Monday) by 4.00 pm.